

Who gains from AI?

Autonomy, capability, and the knowledge economy

Ide & Talamàs (2025)

github.com/gsaco/ai-05-ide

Reading: course PDF, May 20, 2025 (39 pp).
Lean pin: arXiv v11, February 25, 2025 (35 pp).

Two dimensions govern the distribution of gains

Autonomy: what roles can AI take?

Autonomous: independent producer, worker, or solver.

Non-autonomous: solver / co-pilot only.

Capability: what can AI solve?

AI knows solutions up to difficulty
 $a = z_{AI} < 1$.

A permitted role need not be profitable at a low capability.

The question is who earns more *in equilibrium*, after occupations, matches, wages, and compute rents adjust.

Road map: propositions $\rightarrow$ three-type derivation $\rightarrow$ computation $\rightarrow$ Lean boundary.

The agent's problem: allocate time and knowledge

A worker knows z ; a solver knows $s > z$; difficulty $x \sim U[0, 1]$.

$$\underbrace{h n(z)(1 - z)}_{\text{solver time used}} = 1 \quad \implies \quad n(z) = \frac{1}{h(1 - z)}.$$

$$\max_{\text{organization}, z, s} \Pi, \quad \Pi_{HH} = n(z)[s - w(z)] - w(s).$$

- ▶ Humans choose income-maximizing roles; compute owners seek rent r .
- ▶ Competitive entry: active firms earn zero profit; all resource markets clear.
- ▶ Output per worker is s , including problems solved without assistance.

Section 3.1 in both versions. AI replaces the relevant wage by its compute rental rate.

The headline results have important conditions

- ▶ Unit mass of humans, one unit of time each; observable $z \in [0, 1]$ with continuous, strictly positive density g .
- ▶ Uniform independent difficulties, risk neutrality, output price one; competitive firms, no fixed costs, at most two layers.
- ▶ $0 < h < h_0$: no independent human producers before AI.
- ▶ $a \in [0, 1)$; opportunities exceed human-plus-AI production capacity.
- ▶ Compute abundant relative to human time. For Proposition 6, compare the *same* capability and compute supply.

The top-winner claim needs both $h < h_0$ and $a < 1$. Perfect AI is a different boundary case.

Sections 3.1, 3.3, 5.2, and 6. Exact sufficient compute bound in the appendix slide.

Proposition 5: bottom winners require capability

Define autonomous-AI winners relative to no AI:

$$B = \{z \in [0, a] : w^A(z) > w(z)\}, \quad T = \{z \in [a, 1] : w^A(z) > w(z)\}.$$

Under the stated model conditions

$$B \neq \emptyset \iff a > \bar{a}, \quad \bar{a} \in \text{int } W; \quad T \neq \emptyset \quad \forall a \in [0, 1).$$

W is the **pre-AI worker set**. Winners lie at the extremes; humans with knowledge a lose.

Even “basic” AI can help the bottom: $\bar{a}$ lies *inside* the worker region. Autonomy alone does not settle the sign.

Proposition 5: v11 p.24; course PDF p.25. w^A denotes the paper's w^* .

Proposition 6: free co-pilots still need to be useful

$$r^N = 0, \quad \begin{cases} a \leq w(0) : & \text{AI unused; pre-AI wages and occupations unchanged,} \\ a > w(0) : & \text{lowest-knowledge humans use AI advice.} \end{cases}$$

- ▶ Near the bottom: $w^N(z) \geq \max\{w(z), w^A(z)\}$, **strict** if $a > w(0)$.
- ▶ Some $z > 0$ has $w^N(z) \leq w(z)$, **strict** if $a > w(0)$.
- ▶ Near the top: $w^N(z) \leq w^A(z)$, strict away from $z = 1$.
- ▶ Output: $Y^A > Y^N$. Efficiency is relative to the permitted technology.

The thresholds $w(0)$ and $\bar{a}$ answer different questions. Neither can be dropped.

Why the bottom and top respond differently

Bottom: matching versus income share

$$w(z) = s - h(1 - z)w(s).$$

A worse solver reduces output; a cheaper solver lets the worker retain more.

Capability changes the balance.

Autonomous compute has an outside option: $r^A = a$. Co-pilot-only compute can remain idle: $r^N = 0$.

Wage formulas apply to their respective active roles; they are not global wage formulas.

Top: scale up scarce expertise

$$w^A(s) = \frac{s - a}{h(1 - a)}$$

when supervising AI workers. Low-cost routine work increases the return to human problem-solving.

A three-type economy we can solve by hand

$$(z_L, z_M, z_H) = (0, \frac{1}{2}, 1), \quad (m_L, m_M, m_H) = (\frac{11}{20}, \frac{3}{10}, \frac{3}{20}), \quad h = \frac{1}{2}, \quad \mu = 2.$$

Let wages be (u, v, t) . Three active human teams imply

$$u + \frac{1}{2}v = \frac{1}{2}, \quad u + \frac{1}{2}t = 1, \quad v + \frac{1}{4}t = 1.$$

$$t - v = 1 \quad \Rightarrow \quad \boxed{(u, v, t) = (\frac{1}{5}, \frac{3}{5}, \frac{8}{5})}.$$

Worker masses $(L \rightarrow M, L \rightarrow H, M \rightarrow H) = (.32, .23, .14)$ clear all three markets; $Y_0 = .53$.

This is a finite-type analogue. It changes the source's density assumption and does not inherit its continuum theorems.

Autonomous AI: derive an equilibrium branch

Take $0 < a < \frac{1}{3}$. Low humans work with middle solvers; the remaining middle humans and all high humans supervise AI.

$$r^A = a, \quad t = \frac{1-a}{(1-a)/2} = 2, \quad v = \frac{\frac{1}{2} - a}{(1-a)/2} = \frac{1-2a}{1-a}.$$

$$u = \frac{1}{2} - \frac{1}{2}v = \boxed{\frac{a}{2(1-a)}}.$$

- ▶ Low workers use $.55/2 = .275$ middle solvers, leaving $.025$.
- ▶ AI workers use $7/[20(1-a)] < 2$ compute; the rest produces independently.
- ▶ The other no-entry inequalities hold; $v > \frac{1}{2}$ on this branch.

An allocation plus a feasible no-profit price vector is the equilibrium certificate.

The threshold is visible in three lines of algebra

$$w_L^A > w_L \iff \frac{a}{2(1-a)} > \frac{1}{5}$$

$$\iff 5a > 2(1-a) \iff \boxed{a > \frac{2}{7}}.$$

Multiplication preserves the inequality because $1 - a > 0$.

Same autonomy; different capability	$a = 1/5$	$a = 3/10$
Lowest wage	$1/8 < 1/5$	$3/14 > 1/5$
Verdict	loses	gains

Both values lie in $0 < a < 1/3$. Equality at $a = 2/7$ is not a strict gain. This is not the continuum's numerical threshold.

Co-pilot-only AI: solve the same economy at $a = .4$

Spare compute is idle, so $r^N = 0$. Low humans using AI earn $u = a = .4$.

$$u + \frac{1}{2}t = 1 \Rightarrow t = 1.2, \quad v + \frac{1}{4}t = 1 \Rightarrow v = .7.$$

Allocation

- ▶ .30 middle workers use high solvers.
- ▶ .15 low workers use high solvers.
- ▶ .40 low workers use AI advice.

Income comparison

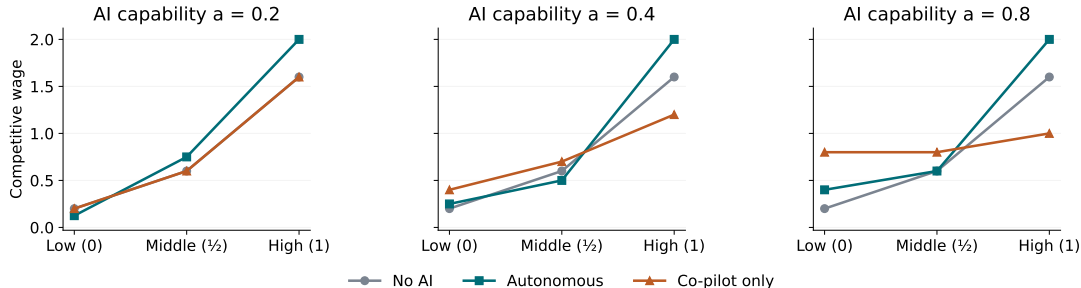
$$(.2, .6, 1.6) \longrightarrow (.4, .7, 1.2).$$

Bottom and middle gain; top loses.

$Y^N = .61$; all of it is labor income.

At $a = .2$, the reported non-autonomous equilibrium remains the pre-AI one: the capability boundary matters.

Computation checks prices and quantities together



Seven equilibria: no AI and two regimes at three capabilities.

Checked resource feasibility, no profitable entry, complementary slackness, primal–dual equality, wage uniqueness, and $Y = L + \mu r$.

Reproduce: `python3 analysis/discrete_model.py`. Lines connect discrete points only; they do not represent a continuum.

Higher output is not the same as higher wages for all

Regime at $a = .4$	Labor L	Capital μr	Output Y
No AI	.5300	0	.5300
Co-pilot only	.6100	0	.6100
Autonomous	.5875	.8000	1.3875

Every co-pilot-only allocation is feasible with autonomy. For $a > 0$, using its idle compute $\delta > 0$ independently adds $a\delta > 0$ output.

Autonomy raises total output here while total human wage income is lower than under co-pilots. Track compute-owner income separately.

At $a = 0$, this strictness argument fails; AI workers assisted by humans supply the separate mechanism.

Lean checks the zero-profit wage calculation

Paper equation (Section 3.1):

$$n(z) = \frac{1}{h(1-z)}, \quad n(z)(a-w) - r = 0 \implies w = a - h(1-z)r.$$

```
def ZeroProfitWageSpec : Prop :=  
  ∀ h z a w r : ℝ, 0 < h → h < 1 → 0 ≤ z → z < 1 →  
    topAutomatedProfit h z a w r = 0 → w = a - h * (1 - z) * r  
  
theorem zeroProfitWage : ZeroProfitWageSpec := by  
  intro h z a w r hh _ _ hz hprofit  
  have hhne : h ≠ 0 := ne_of_gt hh  
  have hzne : 1 - z ≠ 0 := ne_of_gt (sub_pos.mpr hz)  
  unfold topAutomatedProfit span at hprofit  
  field_simp [hhne, hzne] at hprofit  
  nlinarith
```

Interpretation: real scalars encode wages, rent, knowledge, and helping cost. The have lines justify clearing denominators; nlinarith closes the algebra.

FirmAlgebra.zeroProfitWage: focused Lean build PASS. This verifies the conditional algebraic implication.

The translation boundary is part of the result

- ▶ **Checked:** zero-profit wage algebra and its $r = a / r = 0$ specializations; the finite branch's strict threshold $a > 2/7$.
- ▶ **Scalar generalization:** the displayed theorem lets $a, w, r \in \mathbb{R}$. Economic use also requires admissible capability, a suitable worker, and an active team.
- ▶ **Still unproved in Lean:** equilibrium existence, clearing, sorting, uniqueness, and the continuum Propositions 1–6. Neither $r = a$ nor the candidate wage branch is derived from equilibrium in Lean.

Actual run status

Focused support-module build: **PASS**.

Required check `IT25KnowledgeEconomy -fast`: **PASS (exit 0)**.

This checks the build and whitespace; **0/6 source propositions proved**.

A checked support lemma is not a completed paper formalization. See `lean/` and `checks/lean-check.md`.

The claim to challenge on a handwritten page

Genuine photograph still required

Write and check the derivation, then save the photograph as `hand/derivation.jpg`.
Recompile to display it here.

$$\frac{a}{2(1-a)} > \frac{1}{5} \iff a > \frac{2}{7}, \quad 0 < a < \frac{1}{3}.$$

Proposed verdict: reject the autonomy-only gloss. Check the positive denominator, strict inequality, and equilibrium branch yourself.

A typeset equation is not evidence that the student performed a hand derivation.

What the checks teach us

1. **Bottom:** capability crosses a threshold, even with autonomy fixed.
2. **Top:** autonomous AI scales scarce expertise; the source result needs $h < h_0$ and $a < 1$.
3. **Output:** more feasible uses of compute raise output, without making every human better off.
4. **Verification:** an algebraic proof, an equilibrium computation, and a continuum theorem are different achievements.

The defensible conclusion is a two-dimensional taxonomy, with domains and inequalities stated explicitly.

Reference: complete Proposition 6 clauses

The non-autonomous equilibrium is unique, efficient, maximizes labor income within that technology, and $r^N = 0$. If $a \leq w(0)$, AI is unused. If $a > w(0)$,

$$W_a^N \subseteq W_p^N \cup I^N \cup S_p^N; \quad W_a^N, W_p^N, S_p^N \neq \emptyset$$

and only I^N may be empty. In either case:

$$Y^A > Y^N;$$

$$\exists z \in (0, 1] : w^N(z) \leq w(z) \quad (\text{strict if } a > w(0));$$

$$\exists \epsilon > 0 \forall z \in [0, \epsilon) : w^N(z) \geq \max\{w(z), w^A(z)\} \\ (\text{strict if } a > w(0));$$

$$\exists \epsilon > 0 \forall z \in (1 - \epsilon, 1] : w^N(z) \leq w^A(z) \quad (\text{strict if } z \neq 1).$$

Source: v11 p.27 / course p.28. W_a : AI-assisted; W_p : human-assisted; I : independent; S_p : human solvers.

Reference: Lean checks the finite strict threshold

$$0 < a < \frac{1}{3} \implies \left(\frac{1}{5} < \frac{a}{2(1-a)} \iff \frac{2}{7} < a \right).$$

```
def BottomGainSpec : Prop :=  
  ∀ a : ℝ, 0 < a → a < 1 / 3 →  
    (1 / 5 < bottomWage a ↔ 2 / 7 < a)  
  
theorem bottomGain : BottomGainSpec := by  
  intro a ha hupper  
  have hden : 0 < 2 * (1 - a) := by linarith  
  unfold bottomWage  
  rw [lt_div_iff0 hden]  
  constructor <;> intro h <;> linarith
```

hden licenses inequality-preserving division; constructor proves both directions. This is candidate-wage algebra; its equilibrium interpretation is separate.

Discrete.bottomGain: focused Lean build PASS. Original finite example; not Proposition 5 of the source.

Reference: compute, source versions, and reproducibility

A sufficient abundance condition is

$$\mu > \int_0^a h(1-z) dG(z) + \frac{1-G(a)}{h(1-a)}.$$

- ▶ Course PDF: May 20, 2025, 39 pages; arXiv v12 stamp.
- ▶ Formalization pin: arXiv v11, title date February 25, 2025; submission February 24; 35 pages.
- ▶ Proposition 5: v11 p.24 / course p.25. Proposition 6: v11 p.27 / course p.28.
- ▶ Full algebra and computational certificates: `analysis/`. Raw conversation: `prompts.md`. Actual formalization check: `checks/`.
- ▶ Citation: Enrique Ide and Eduard Talamàs (2025), *Artificial Intelligence in the Knowledge Economy*.

arxiv.org/abs/2312.05481v11 github.com/gsaco/ai-05-ide